MT 1003 – Calculus and Analytical Geometry

Homework #4

Points of Non-Differentiability, Monotonicity, and Concavity

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Instructions:

1. Name the submitted file as `Section RollNumber HW4` (for example, `A 26I1234 HW4`).
2. Submit the complete homework as one scanned PDF file.
3. Show the required derivatives, critical numbers, sign charts, and conclusions. Unsupported final answers will not receive full credit.
4. Draw sign charts on a real line and mark all relevant points clearly.
5. For Section B, include the Python code, generated graphs, and your written observations in the same PDF file.

Section A: Analytical Work

I. Points of Non-Differentiability

Q1. For each function, find every point at which it is not differentiable. Where appropriate, write the function piecewise and use the left-hand and right-hand derivatives. Classify each failure as a corner, cusp, vertical tangent, or discontinuity.

(a) $f(x) = |x^2 - 5x + 6|$

(b) $f(x) = |(x+2)^3(x-1)^2|$

(c) $f(x) = |2x+1|^{4/5}$

(d) $f(x) = (x-4)^{1/5}$

Q2. Determine all points at which the following function is differentiable. At each joining point, check continuity first and then compare the one-sided derivatives.

$$g(x) = \begin{cases} x^2 + 2x, & x < 0, \\ 2x, & 0 \le x < 2, \\ (x-2)^2 + 4, & x \ge 2. \end{cases}$$

II. Increasing and Decreasing Intervals

Q3. For each function below:

(i) find all critical numbers;

(ii) construct the sign chart of $f'(x)$;

(iii) state the intervals on which f is increasing or decreasing; and

(iv) identify any local maximum or local minimum using the First Derivative Test.

(a) $f(x) = x^3 + 3x^2 - 9x + 4$

(b) $f(x) = x^4 - 8x^2$

(c) $f(x) = \dfrac{x}{x^2 + 4}$

III. Concavity and Points of Inflection

Q4. For each function below:

(i) find the points where $f''(x) = 0$ or $f''(x)$ does not exist;

(ii) construct the sign chart of $f''(x)$;

(iii) state the intervals of concavity; and

(iv) find all points of inflection, checking continuity where necessary.

(a) $f(x) = 2x^3 - 3x^2 - 12x + 1$

(b) $f(x) = x^4 - 4x^3 + 2$

(c) $f(x) = (x - 1)e^x$

Q5. Complete Analysis. For

$$h(x) = x^3 + 3x^2 - 9x - 4,$$

find its critical numbers, increasing and decreasing intervals, local extrema, intervals of concavity, and point(s) of inflection. Include both the h' and h'' sign charts.

Section B: Derivative Analysis with Python

Q6. Visualizing a Function and Its Derivatives. Run the following program for

$$p(x) = x^4 - 4x^2 + 1.$$

The program plots p, p', and p'' on the same interval.

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-3, 3, 1000)
p = x**4 - 4*x**2 + 1
dp = 4*x**3 - 8*x
d2p = 12*x**2 - 8

fig, axes = plt.subplots(1, 3, figsize=(15, 4))

axes[0].plot(x, p, color="navy")
axes[0].axhline(0, color="black", linewidth=0.7)
axes[0].axvline(0, color="black", linewidth=0.7)
axes[0].set_title("p(x)")
axes[0].grid(True)

axes[1].plot(x, dp, color="darkgreen")
axes[1].axhline(0, color="black", linewidth=0.7)
axes[1].axvline(0, color="black", linewidth=0.7)
axes[1].set_title("p'(x)")
axes[1].grid(True)

axes[2].plot(x, d2p, color="darkred")
axes[2].axhline(0, color="black", linewidth=0.7)
axes[2].axvline(0, color="black", linewidth=0.7)
axes[2].set_title("p''(x)")
axes[2].grid(True)

plt.tight_layout()
plt.show()
```

Using the graphs and exact calculations, answer the following.

(a) Estimate the zeros of $p'(x)$ from the graph, and then find them exactly.

(b) State the intervals on which p is increasing and decreasing.

(c) Identify the local maxima and minima of p.

(d) Estimate the zeros of $p''(x)$ from the graph, and then find them exactly.

(e) State the intervals of concavity and find the points of inflection.

(f) In two or three sentences, explain how the graphs of p', p'', and p support your conclusions.